Research Design I: Fundamentals

Advanced Topics: Methodology, Philosophy, and Research Design

1. From Problem to Research Question

Starting point is a problem or research interest.

Research question: Precise, empirically answerable question. Criteria: clear, relevant, answerable, bounded.

Hypothesis: Precise, testable prediction. Formulate H0 and H1.

2. Exploratory vs. Confirmatory Research

Exploratory (hypothesis-generating): Little prior knowledge. Open questions. Qualitative methods, data mining.

Confirmatory (hypothesis-testing): Prior knowledge exists. Precise hypotheses. Experiments, standardized surveys.

Important: Do not use the same data for both exploratory and confirmatory purposes (p-hacking / HARKing).

3. Cross-Sectional vs. Longitudinal Design

Cross-sectional: Data at one point in time. Cheap, fast. No causal ordering, no intra-individual change.

Longitudinal: Multiple measurement points (panel, trend). Intra-individual change, temporal ordering. Expensive, dropout.

4. Validity as Quality Criterion

Internal validity: Does the study measure the causal effect? Are alternative explanations ruled out?

External validity: Are results generalizable (population, situation, time)?

Construct validity: Does the instrument adequately measure the theoretical construct?

Statistical validity: Are statistical conclusions correct (Type I/II errors, assumptions)?

5. Common Design Errors

Sampling bias: Sample not representative (e.g., only volunteers -> over-motivated).

Confounding: Third variable creates spurious correlation.

Dropout in longitudinal studies: Only survivors measured -> systematic bias.

Ceiling/floor effects: Scale too easy/difficult -> no variance -> no measurable effects.

Common method bias: All variables measured with same method -> artificially inflated relationships.